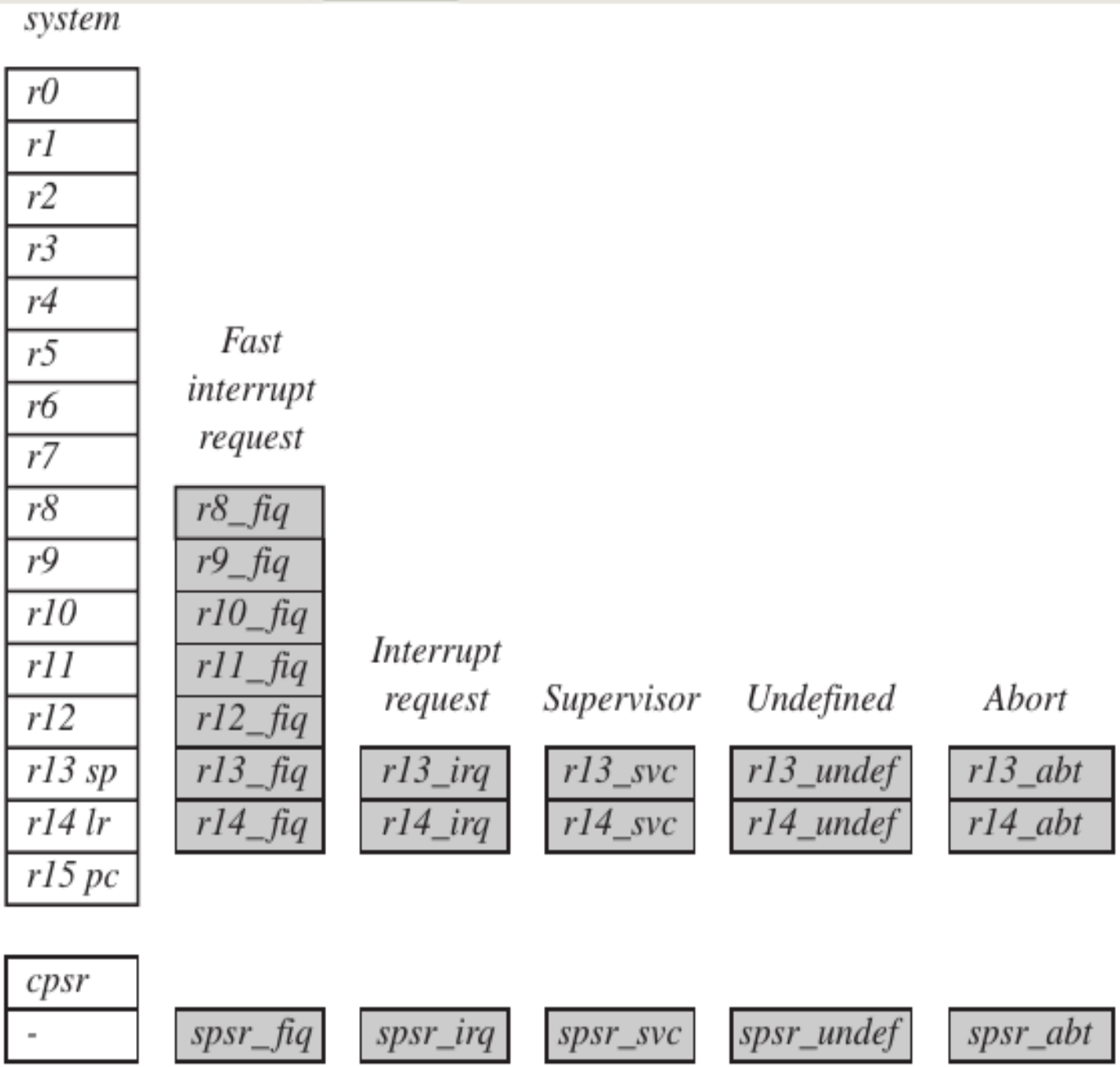
ARM 7 Registers / Programmers Model

System and User	FIQ	Supervisor	Abort	IRQ	Undefined
r0	r0	r0	r0	r0	r0
r1	r1	r1	r1	r1	r1
r2	r2	r2	r2	r2	r2
r3	r3	r3	r3	r3	r3
r4	r4	r4	r4	r4	r4
r5	r5	r5	r5	r5	r5
r6	r6	r6	r6	r6	r6
r7	r7	r7	r7	r7	r7
r8	r8_fiq	r8	r8	r8	r8
r9	r9_fiq	r9	r9	r9	r9
r10	r10_fiq	r10	r10	r10	r10
r11	r11_fiq	r11	r11	r11	r11
r12	r12_fiq	r12	r12	r12	r12
r13	r13_fiq	r13_svc	r13_abt	r13_irq	r13_und
r14	r14_fiq	r14_svc	r14_abt	r14_irq	r14_und
r15 (PC)	r15 (PC)	r15 (PC)	r15 (PC)	r15 (PC)	r15 (PC)

ARM-state program status registers

CPSR	CPSR	CPSR	CPSR	CPSR	CPSR
	SPSR_fiq	SPSR_svc	SPSR_abt	SPSR_irq	SPSR_und


system
r0
r1
r2
r3
r4
r5
r6
r7
r8
r9
r10
r11
r12
r13 sp
r14 lr
r15 pc
cpsr
-
Fast interrupt request
r8_fiq
r9_fiq
r10_fiq
r11_fiq
r12_fiq
r13_fiq
r14_fiq
spsr_fiq
Interrupt request
r13_irq
r14_irq
spsr_irq
Supervisor
r13_svc
r14_svc
spsr_svc
Undefined
r13_undef
r14_undef
spsr_undef
Abort
r13_abt
r14_abt
spsr_abt

1) There are 7 modes of operations.
In each mode, programmer has access to 16 GPRS (R0 – R15) and a CPSR (Flag Reg).
Additionally a Saved PSR (SPSR) is also available in all modes except User/ System mode.
All registers are of 32 bits. In all modes the three registers have common functions.

2) **R15: PC (Program Counter)**
PC gives a 32-bit address of the instruction to be fetched.
This address is put on the address bus and the instruction is fetched from the memory through the 32-bit data bus. PC is periodically incremented after every instruction is fetched.
Since all instructions are of 32-bits and are stored at 32-bit aligned locations (multiples of 4), the lowest two bits of PC are insignificant (XX).

3) **R14: LR (Link Register)**
It stores the return address (value of PC) when we perform a BL instruction (Branch with link, like a CALL instruction). While returning back to main program the value of LR will be put back into PC.

4) **R13: SP (Stack Pointer)**
Holds the address of the top of stack.

5) Additionally there are some **Banked Registers** that are available in specific operating modes.

6) **FIQ mode:**
R8 – R14 provide a fresh set of GPRs.
This eliminates the need to store the original values of these registers elsewhere and hence makes execution of the ISR faster.

7) All the other privileged modes get their own versions of **SP and LR (R13 and R14).**

8) This gives a total of **37 registers all 32-bits each.**

1) **User Mode**

This is the normal mode in which all user programs are executed.
It is the only non-privileged mode.
It has limited access to memory, I/O components and flags.
All the other modes can be entered through different kinds of exceptions (interrupts).

2) **Fast Interrupt Request Mode**

This mode is entered when a high priority interrupt occurs through nFIQ pin.
As this mode is used for high priority interrupts, the ISR should be executed with minimum latency (delay). Normally when we begin an ISR the original value of all GPRs of the parent program must be saved and before completion of the ISR these values must be restored into the GPRs for proper continuation of the main program. This wastes a lot of time and therefore increases Interrupt Latency. But when a fast Interrupt request occurs, a new copy of R8-R14 are used. This completely eliminates the need for storing and restoring these registers and hence executes the ISR much faster. Therefore it is called Fast Interrupt Mode. By default nested interrupts are disabled in FIQ mode.

3) **Interrupt Request Mode**

This is a normal Interrupt Mode and is invoked when a low priority interrupt occurs on the nIRO pin. Here Interrupt latency is mode and nested interrupts are allowed.

4) **Supervisor Mode**

ARM7 enters this mode on reset.
It is used to execute the BIOS program (Booting program).
This mode can also be invoked by the programmer by SWI (Software Interrupt).

5) **Abort Mode**

This mode is entered when an unsuccessful attempt is made to access a memory location.
Due to protection mechanism some locations are not accessible in some of the modes.
When an attempt is made to access such a location, the processor eneters Abort mode and the program that tried to access this location is aborted.

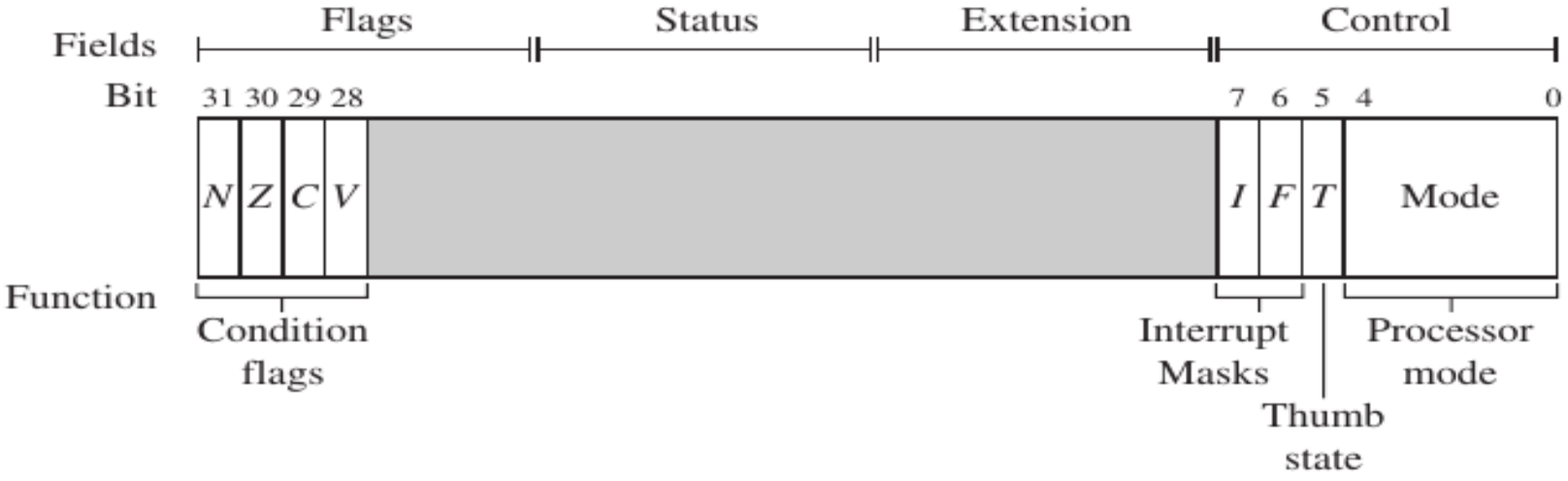
6) **Undefined Mode**

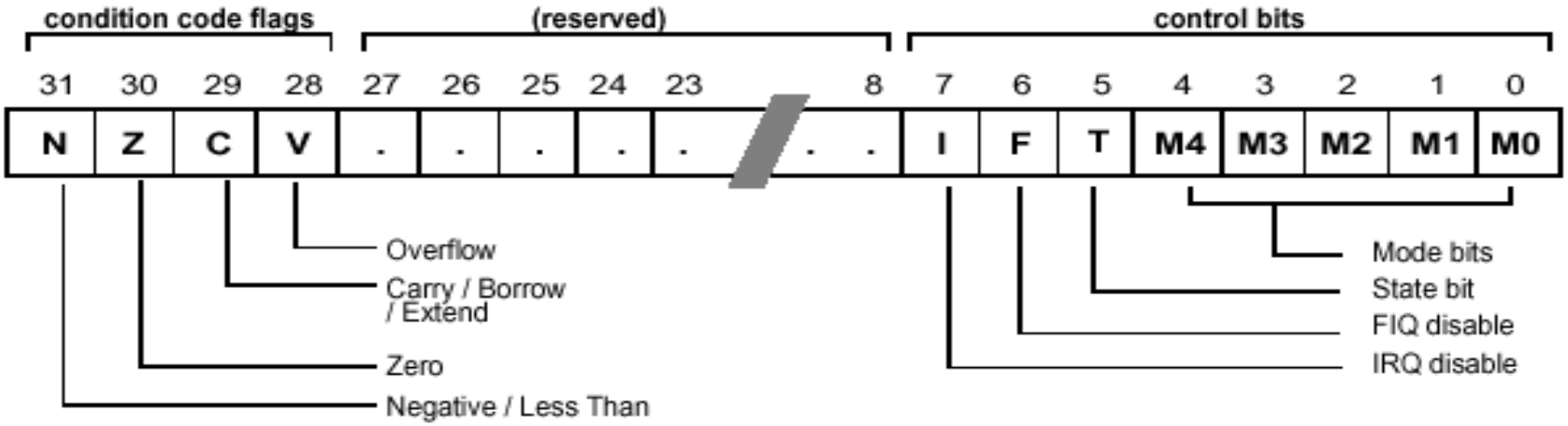
This mode is used to enter undefined instructions.
This generally occurs when a co-processor instruction is encountered but a co-processor is not present in the system. The processor then enters Undefined mode and the program attempting that instruction is terminated.

7) **System Mode**

This is a Privileged version of the User Mode. The User program can invoke this mode to gain full control over CPSR and Memory. Needless to say, this mode should be used very discretely by the programmer, else the entire protection mechanism may be compromised.

R16 is state register/CPSR(Current Program Status Register)





Mode	Abbreviation	Privileged	Bits [4:0]
Abort	abt	Yes	10111
Fast Interrupt	fiq	Yes	10001
Interrupt request	irq	Yes	10010
Supervisor	svc	Yes	10011
System	sys	Yes	11111
Undefined	und	Yes	11011
User	usr	No	10000

Thumb Mode: What is Thumb?

- 16-bit instruction, dynamically decompressed into ARM Instruction before execution
- Fewer directly accessible GPRs (Only R0-R7)
- The CPU can be switched between ARM ISA mode and Thumb ISA mode dynamically
- If ‘T’ bit of CPSR is set, the processor interprets the instruction as 16-bit Thumb instruction

Thumb Mode: What is Thumb?

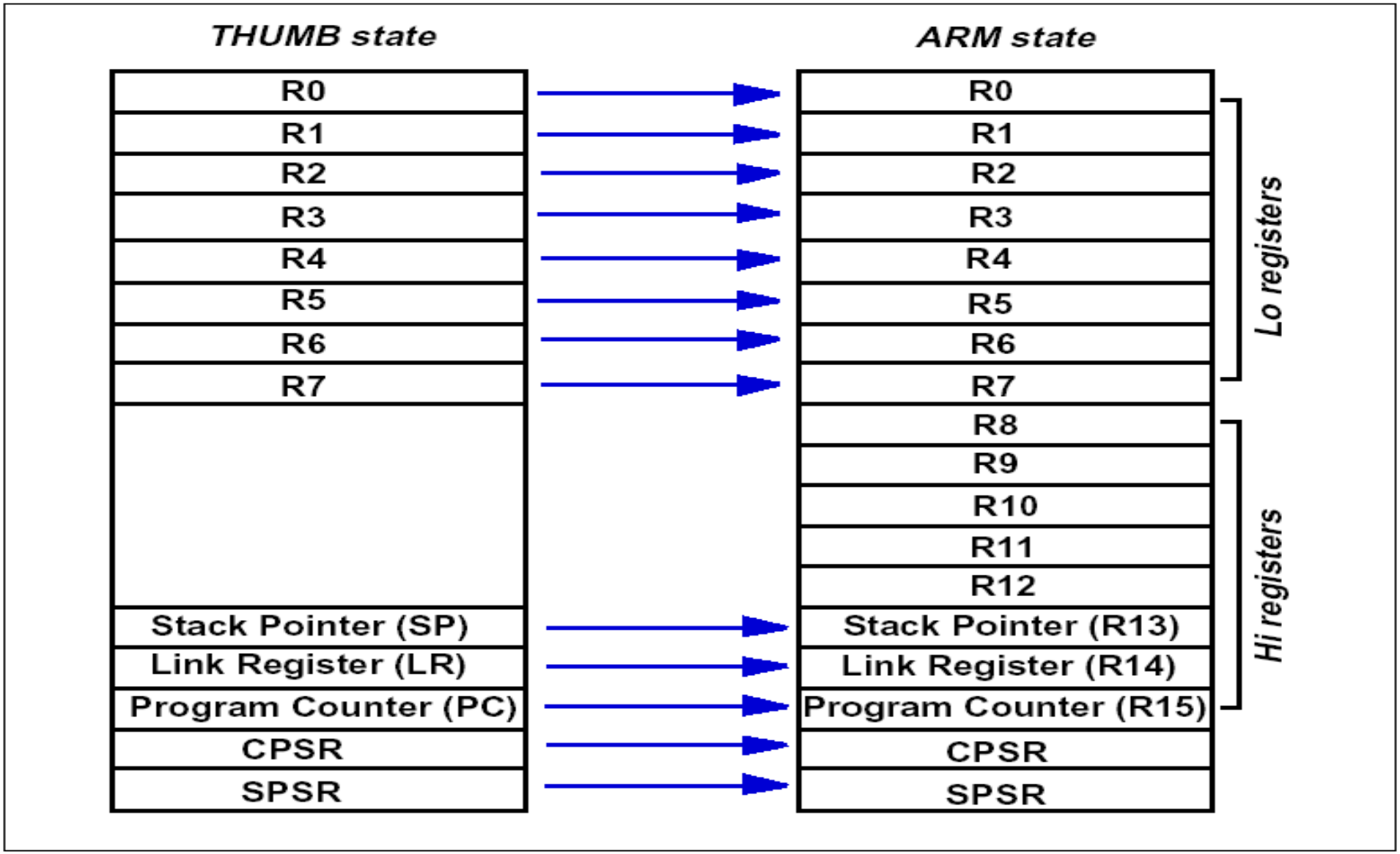


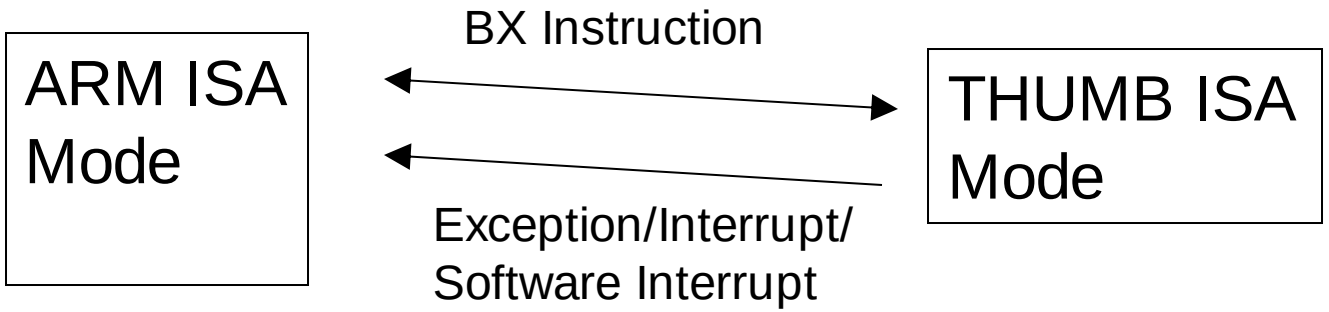
Figure 3-5: Mapping of THUMB state registers onto ARM state registers

Why Thumb?

- Addresses higher code density by packing common instructions into 16-bit
  - Reduces bus width and activities
  - Datapath remain unchanged, thus still 32-bit
- In an typical application, performance-sensitive code can be in ARM ISA, while other code in Thumb
  - Denser code, less performance impact

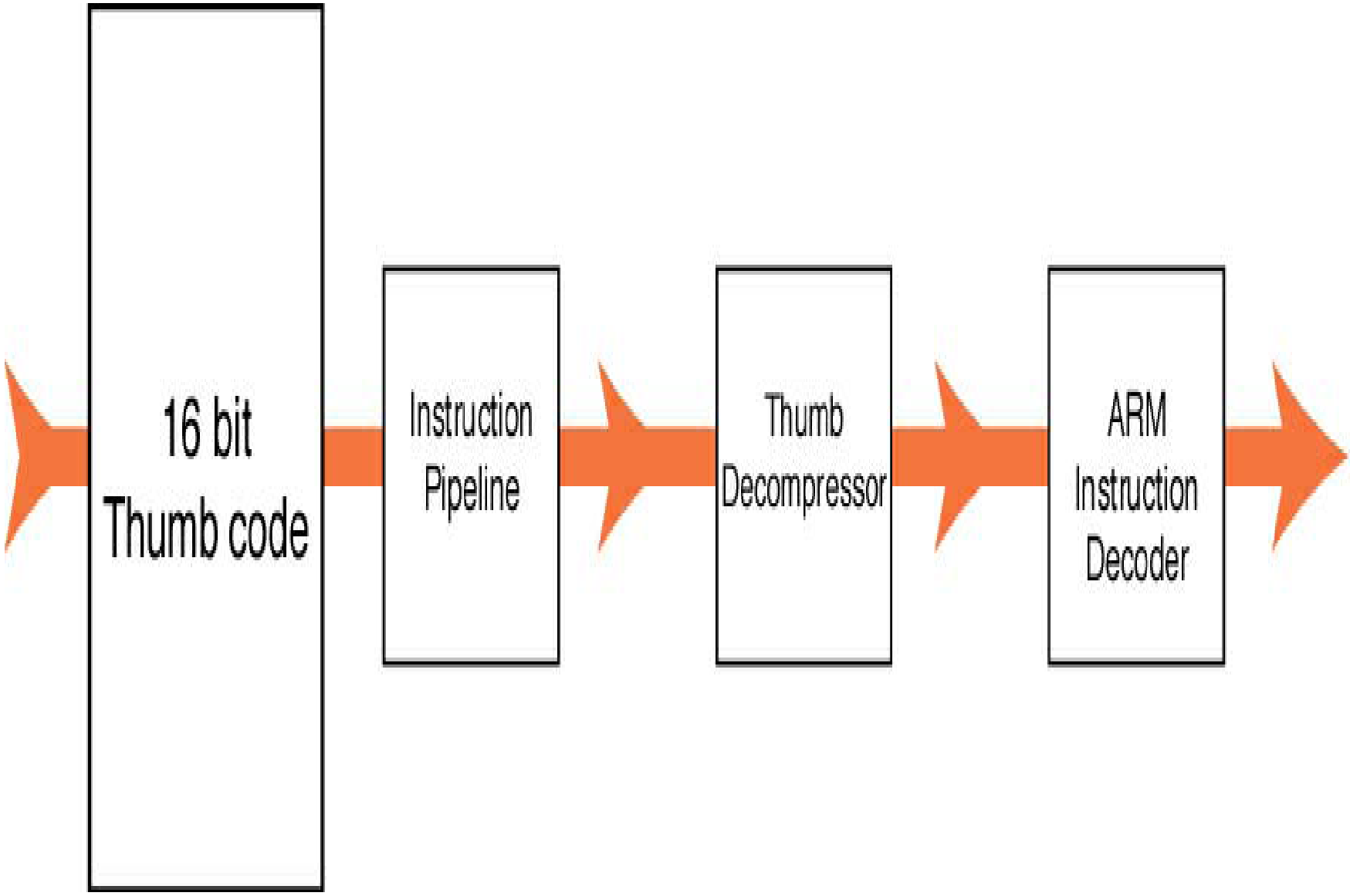
ARM/THUMB Mode switch

- Switch to ARM ISA mode automatically on exception
- Can also be switch by BX (Branch and Exchange) Instruction



Thumb/ARM differences

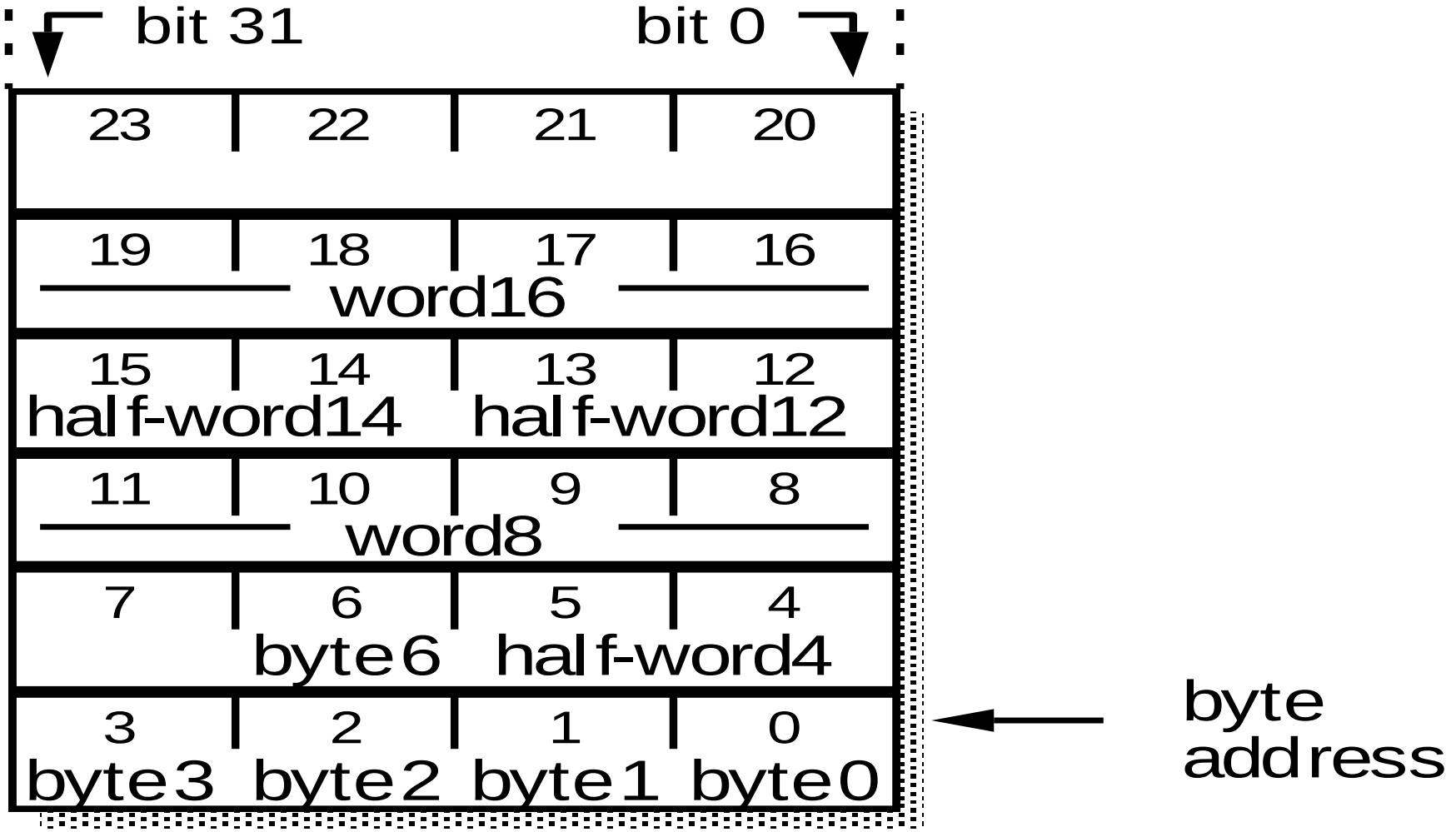
- ALL ARM instructions are conditional
- Most Thumb instructions are unconditional
- ARM are 32 bit instructions
- THUMB are 16-bit compressed instructions


16 bit
Thumb code
Instruction
Pipeline
Thumb
Decompressor
ARM
Instruction
Decoder

The Memory System

- 32 bit Address Bus Memory ranges from 0 up to 2^32 –1
- Data items may be 8-bit (B), 16-bit (HW), or 32-bit (W)
- Words are always aligned at 4-byte boundaries
- Half Words are aligned on even byte boundaries

ARM Memory Organization



I/O System

- I/O peripherals are Memory-Mapped devices with Interrupt Support (IRQ or FIQ).
- Devices registers appear as addressable locations within ARM's Memory Map.
- Writing into or reading from these is done using load-store instructions.